

# Companion to Bird & de Moor, Algebra of Programming

## §6. Recursive Programs

### §6.1. Digits of a number

(6.1a)

$val^\circ = embed^\circ wrap \cup op^\circ -xDigit(val^\circ)snoc$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(a)

(b)

(c)

(d)

(e)

|     |                                                                    |                                                                                                                                           |
|-----|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $val^\circ$                                                        | $val$ undoes $wrap$ or $snoc$ , reads the front with $val$ and combines by $embed$ or $op$ , so its converse runs that backwards          |
| (b) | $([wrap, snoc]^\circ Digit^{+-} (-xDigit(val)) [embed, op])^\circ$ | undoing $wrap$ or $snoc$ first gives two cases: a decimal and its last digit read by $val$ then $op$ , or one digit read by $embed$       |
| (c) | $(wrap^\circ embed)^\circ \cup (snoc^\circ -xDigit(val) op)^\circ$ | the converse of each case is its steps undone in reverse: undo $op$ , undo reading the front, then $snoc$ ; or undo $embed$ , then $wrap$ |
| (d) | $embed^\circ wrap \cup op^\circ -xDigit(val)^\circ snoc$           | undoing the read of the front while keeping the last digit is $val^\circ$ on the front alone                                              |
| (e) | $embed^\circ wrap \cup op^\circ -xDigit(val^\circ) snoc$           |                                                                                                                                           |

(6.1b)

|                                                                                                       |
|-------------------------------------------------------------------------------------------------------|
| $op^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$                                       |
| $op^\circ$ splits a number into its quotient and remainder by 10                                      |
| $(\exists p. 0 < \pi_1 p \wedge op^\circ(m, p)) \iff 10 \leq m$                                       |
| $op^\circ$ gives a pair with a nonzero first component exactly at the numbers with two or more digits |
| $(\exists d. embed^\circ(m, d)) \iff 0 < m \wedge m < 10$                                             |
| $embed^\circ$ gives a digit exactly at the one-digit numbers                                          |

## §6.2. Least fixed points

(6.2a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$ .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$ .

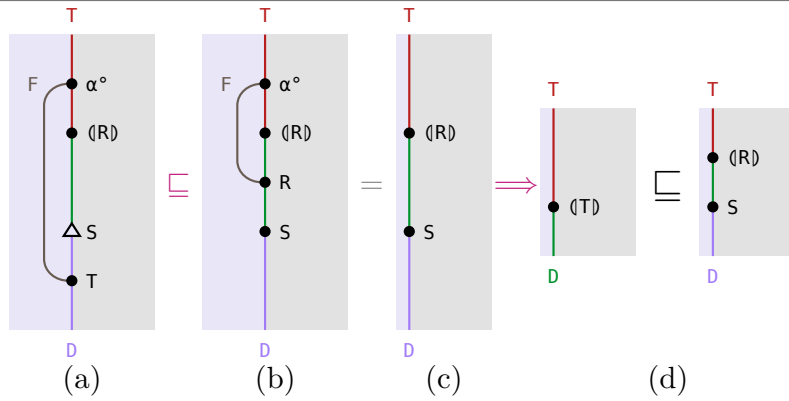
(6.2b)

| the law                                                                                    | what it says                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §Section 6.3 and every chapter after it uses                                      |
| $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |
| $\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$<br>(6.2)   | a relation $X$ that contains one unfolding of the fold's recursion at $X$ contains the fold                                                                                     |
| $X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$<br>(6.3)   | a relation $X$ contained in one unfolding of the fold's recursion at $X$ is contained in the fold                                                                               |

$$F(S)T \sqsubseteq RS \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2c)

if  $S$  followed by  $R$  absorbs  $T$  after  $F(S)$ , the fold of  $T$  is below the fold of  $R$  followed by  $S$

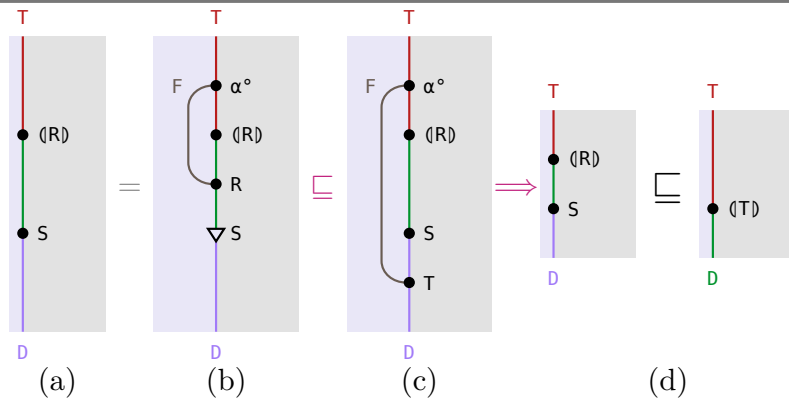


|     |                                                                                                                                       |                                                                                                                                                                                                     |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\alpha^\circ F(\langle R \rangle) F(S) T$                                                                                            | $F(S) T \sqsubseteq RS$                                                                                                                                                                             |
| (b) | $\alpha^\circ F(\langle R \rangle) RS$                                                                                                | $F(\langle R \rangle) R = \alpha^\circ \langle R \rangle, \alpha^\circ \alpha = 1 \triangleq \text{reduce}$                                                                                         |
| (c) | $\langle R \rangle S$                                                                                                                 | $\langle R \rangle S$ is a prefix point of $\alpha^\circ F(-) T$ , and $\langle T \rangle$ is the least one — (6.2) $\alpha^\circ F(X) T \sqsubseteq X \Rightarrow \langle T \rangle \sqsubseteq X$ |
| (d) | $\alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$ |                                                                                                                                                                                                     |

$$R S \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$$

(6.2d)

if  $R$  followed by  $S$  is absorbed by  $F(S)$  followed by  $T$ , the fold of  $R$  followed by  $S$  is below the fold of  $T$



|     |                                                                                                                                       |                                                                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle R \rangle S$                                                                                                                 | $\alpha^\circ \langle R \rangle = F(\langle R \rangle) R, \alpha^\circ \alpha = 1 \triangleq \text{reduce}$                                                                                             |
| (b) | $\alpha^\circ F(\langle R \rangle) RS$                                                                                                | $RS \sqsubseteq F(S) T$                                                                                                                                                                                 |
| (c) | $\alpha^\circ F(\langle R \rangle S) T$                                                                                               | $\langle R \rangle S$ is a postfix point of $\alpha^\circ F(-) T$ , and $\langle T \rangle$ is the greatest one — (6.3) $X \sqsubseteq \alpha^\circ F(X) T \Rightarrow X \sqsubseteq \langle T \rangle$ |
| (d) | $\langle R \rangle S \sqsubseteq \alpha^\circ F(\langle R \rangle S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$ |                                                                                                                                                                                                         |

## §6.3. Hylomorphisms

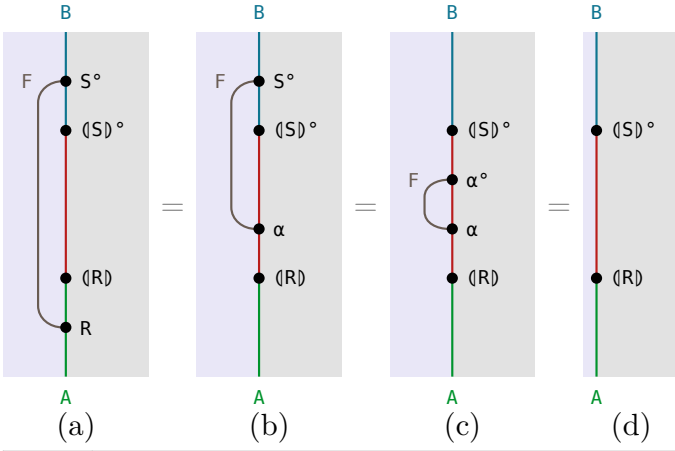
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : S^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

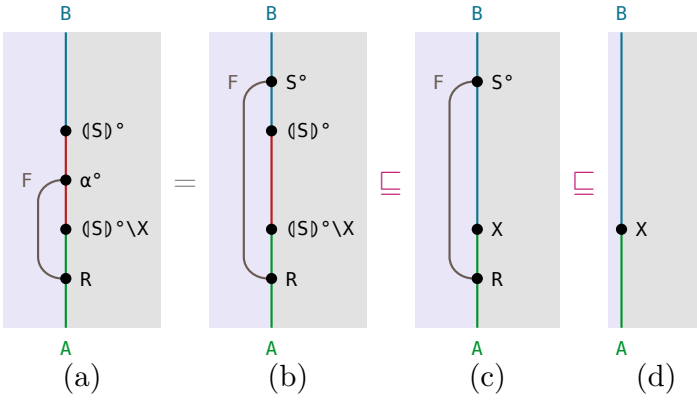
hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term  $S^\circ$  represents the decomposition stage,  $F((\mathbb{S})^\circ(\mathbb{R}))$  the stage of solving the subproblems recursively, and  $R$  the recombination stage;  $R : FA \rightarrow A$ ,  $S : FB \rightarrow B$ ,  $\alpha : FT \rightarrow T$  initial



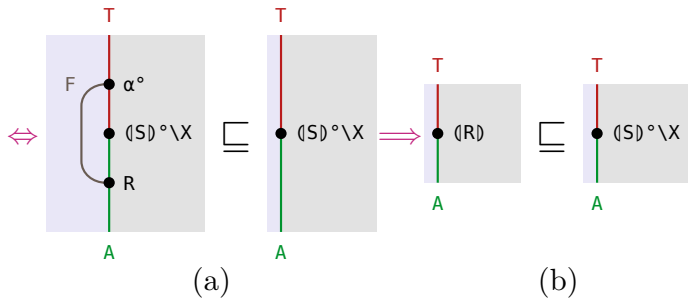
|     |                                                    |                                                                                                         |
|-----|----------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| (a) | $S^\circ F((\mathbb{S})^\circ)F((\mathbb{R}))R$    | $F((\mathbb{R}))R = \alpha(\mathbb{R}) \triangleq \text{reduce}$                                        |
| (b) | $S^\circ F((\mathbb{S})^\circ)\alpha(\mathbb{R})$  | $(\mathbb{S})^\circ\alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$ |
| (c) | $(\mathbb{S})^\circ\alpha^\circ\alpha(\mathbb{R})$ | $\alpha \text{ iso}$                                                                                    |
| (d) | $(\mathbb{S})^\circ(\mathbb{R})$                   |                                                                                                         |

$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism  $(\mathbb{S})^\circ(\mathbb{R})$  is included in  $X$  if  $X$  satisfies the associated recursion inequation



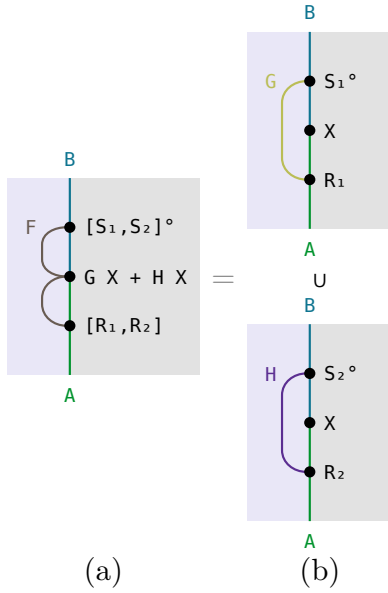
|     |                                                                     |                                                                                      |
|-----|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $(\mathbb{S})^\circ\alpha^\circ F((\mathbb{S})^\circ\backslash X)R$ | $(\mathbb{S})^\circ\alpha^\circ = S^\circ F((\mathbb{S})^\circ)$                     |
| (b) | $S^\circ F((\mathbb{S})^\circ((\mathbb{S})^\circ\backslash X))R$    | $(\mathbb{S})^\circ((\mathbb{S})^\circ\backslash X) \sqsubseteq X \text{ — adj-all}$ |
| (c) | $S^\circ F(X)R$                                                     | $S^\circ F(X)R \sqsubseteq X$                                                        |
| (d) | $X$                                                                 | $S \cdot \dashv S \backslash \text{ — adj-all}$                                      |



|     |                                                                                                                                   |                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| (a) | $S^\circ F(X)R \sqsubseteq X \implies \alpha^\circ F((\mathbb{S})^\circ\backslash X)R \sqsubseteq (\mathbb{S})^\circ\backslash X$ | $(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq \text{reduce, (6.2b), adj-all}$ |
| (b) | $S^\circ F(X)R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ\backslash X$                                    |                                                                                         |

$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \circ ([R_1, R_2])^\circ) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

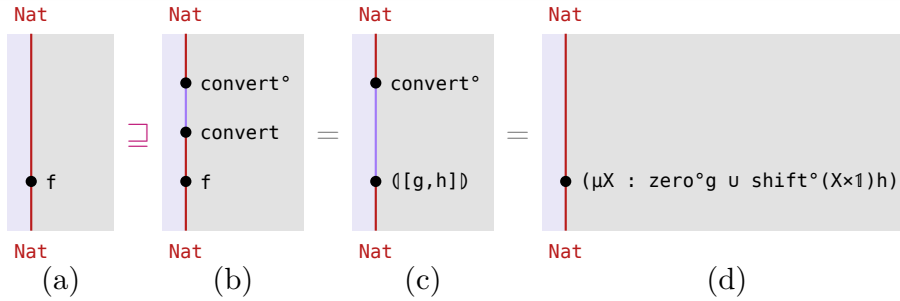


|     |                                              |                                                                                                                                                                                             |
|-----|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[S_1, S_2]^\circ (G(X)+H(X)) [R_1, R_2]$    | $[S, T] = (S+T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$ ; the + stands for the codiagonals $\nabla^\circ, \nabla$ , which are not drawn |
| (b) | $S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$ | $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct                                                                                                                                          |

## §6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1)h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1)h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



|     |                                                                       |                                                               |
|-----|-----------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $f$                                                                   | convert simple                                                |
| (b) | $\text{convert}^\circ \text{convert } f$                              | fusion: $\text{zero } f = g, \text{shift } f = (f \times 1)h$ |
| (c) | $\text{convert}^\circ ([g, h])^\circ$                                 | Corollary 6.1, (6.3b)                                         |
| (d) | $(\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1)h)$ |                                                               |

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

$\text{zero } \text{exp}(a) = \text{one}$

$a$  to the power  $0$  is  $1$

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

$a$  to the power  $2n+d$  is  $\text{op}(a)$  of  $a$  to the power  $n$  and  $d$

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

$\text{zero } \text{mod}(b) = \text{zero}$

$0 \bmod b$  is  $0$

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

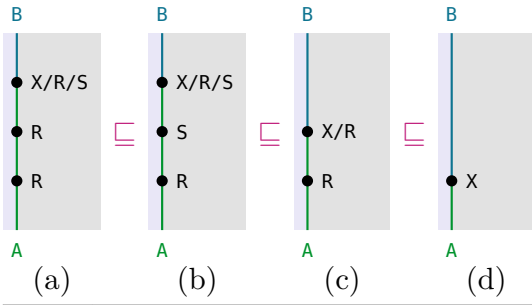
$(2a+d) \bmod b$  is  $\text{op}(b)$  of  $a \bmod b$  and  $d$

## §6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \sqsubseteq SR \Rightarrow \text{inductive}(R)$$

(6.5a)

$R$  is inductive when  $R \sqsubseteq SR$  for an inductive  $S$



|     |             |                    |
|-----|-------------|--------------------|
| (a) | $(X/R/S)RR$ | $R \sqsubseteq SR$ |
| (b) | $(X/R/S)SR$ | division by $S$    |
| (c) | $(X/R)R$    | division by $R$    |
| (d) | $X$         |                    |

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of  $F+G$  is a member of whichever summand the value is in

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of  $F \times G$  is a member of either component

$$\text{inductive}(S \text{ member}) \wedge X = SF(X)R \wedge Y = SF(Y)R \Rightarrow X = Y$$

(6.5c)

when  $S \text{ member}$  is inductive, two solutions  $X, Y$  of  $X = SF(X)R$  are equal

$$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge SF(X)R \in X \Rightarrow \text{Entire}(X)$$

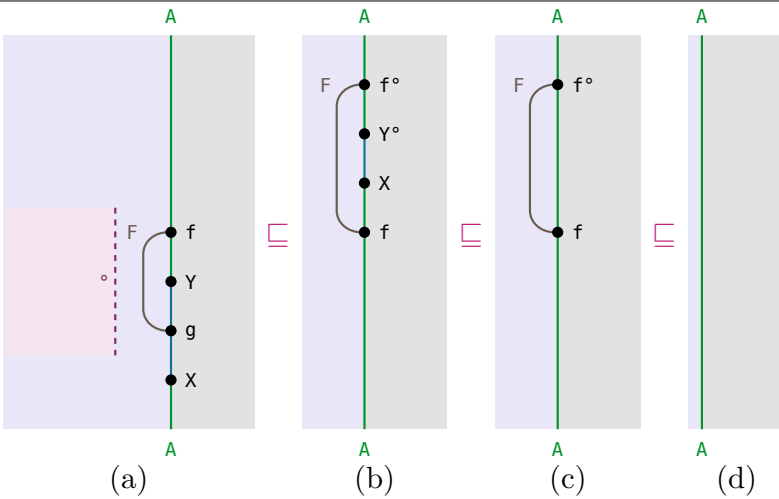
(6.5d)

when  $S \text{ member}$  is inductive and  $S, R$  are entire, a pre-fixed point of  $X \mapsto SF(X)R$  is entire

$$\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

when  $g \text{ member}$  is inductive and  $f, g$  are maps, the solution of  $X = gF(X)f$  is a map

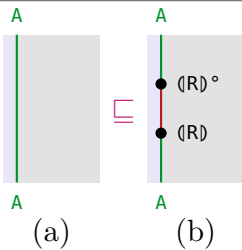


|     |                         |                                |
|-----|-------------------------|--------------------------------|
| (a) | $(gF(Y)f)^\circ X$      | $X = gF(X)f, g \text{ simple}$ |
| (b) | $f^\circ F(Y^\circ X)f$ | $Y^\circ X \in \mathbb{I}$     |
| (c) | $f^\circ f$             | $f \text{ simple}$             |
| (d) | $1$                     |                                |

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \in R^\circ R \Rightarrow 1 \in (R)^\circ (R)$$

(6.5f)

when  $R^\circ \text{ member}$  is inductive and  $R$  is surjective, the fold  $(R)$  is surjective

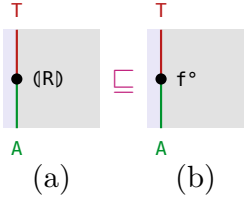


|     |                 |                         |
|-----|-----------------|-------------------------|
| (a) | $1$             | (6.5d) at $S = R^\circ$ |
| (b) | $(R)^\circ (R)$ |                         |

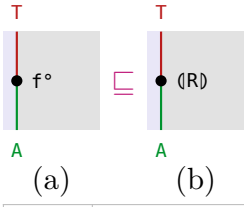
$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = \langle R \rangle$$

(6.5g)

a map  $f$  with  $Rf \sqsubseteq F(f)\alpha$ , for a surjective  $R$ , has the fold  $\langle R \rangle$  as its converse



|     |                     |                                                    |
|-----|---------------------|----------------------------------------------------|
| (a) | $\langle R \rangle$ | shunting $f$ , fusion, $Rf \sqsubseteq F(f)\alpha$ |
| (b) | $f^\circ$           |                                                    |

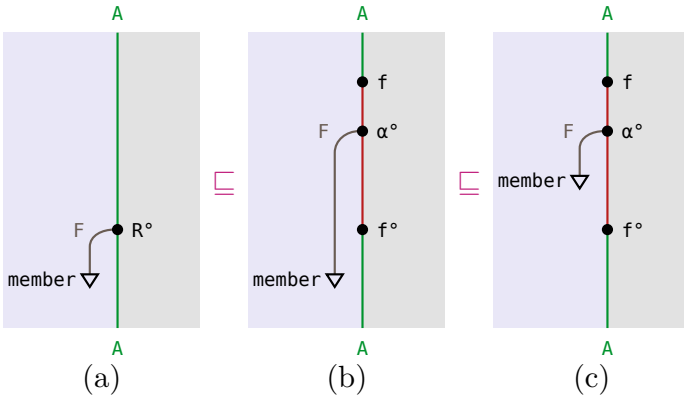


|     |                     |                                                |
|-----|---------------------|------------------------------------------------|
| (a) | $f^\circ$           | (6.5f) and $\langle R \rangle f \sqsubseteq 1$ |
| (b) | $\langle R \rangle$ |                                                |

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \Rightarrow R^\circ \text{member} \sqsubseteq f\alpha^\circ \text{member } f^\circ$$

(6.5h)

$R^\circ \text{member}$  is below  $\alpha^\circ \text{member}$  conjugated by  $f$ , so it is inductive when  $\alpha^\circ \text{member}$  is



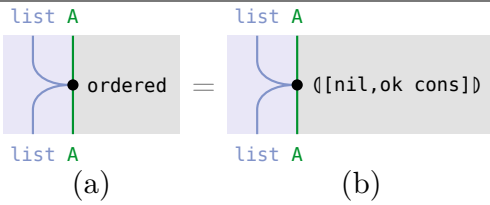
|     |                                          |                                        |
|-----|------------------------------------------|----------------------------------------|
| (a) | $R^\circ \text{member}$                  | $Rf \sqsubseteq F(f)\alpha$ , shunting |
| (b) | $f\alpha^\circ F(f^\circ) \text{member}$ | $\text{member}$ lax natural            |
| (c) | $f\alpha^\circ \text{member } f^\circ$   |                                        |

## §6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

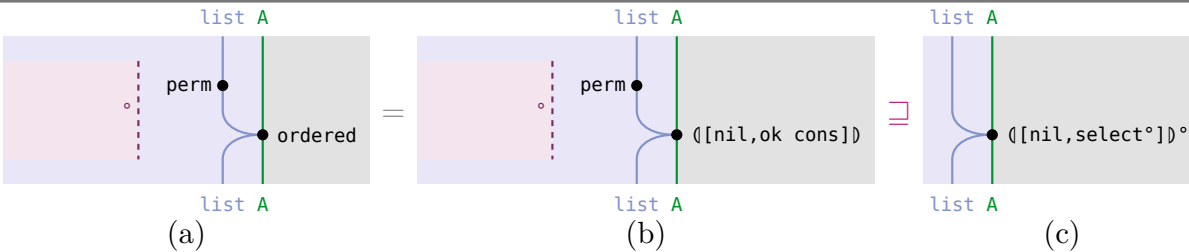


|     |                                                |                 |
|-----|------------------------------------------------|-----------------|
| (a) | $\text{ordered}$                               | fold uniqueness |
| (b) | $\langle [\text{nil}, \text{ok cons}] \rangle$ |                 |

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

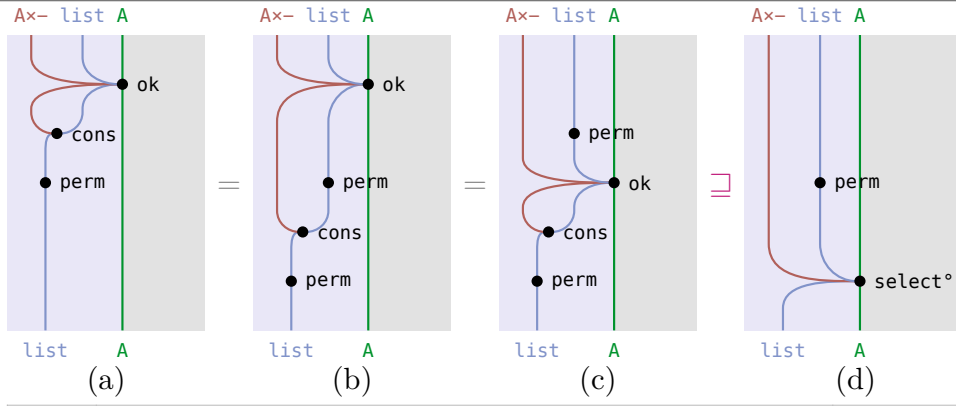
every output of unfolding the input by **select** is a sorted permutation of it



|     |                                                                    |                           |
|-----|--------------------------------------------------------------------|---------------------------|
| (a) | $(\text{ordered perm})^\circ$                                      | (6.6a)                    |
| (b) | $(\langle [\text{nil}, \text{ok cons}] \rangle \text{perm})^\circ$ | fusion (6.4) under (6.6c) |
| (c) | $\langle [\text{nil}, \text{select}^\circ] \rangle^\circ$          |                           |

$$\text{select}^\circ \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \text{ok cons perm} \quad (6.6c)$$

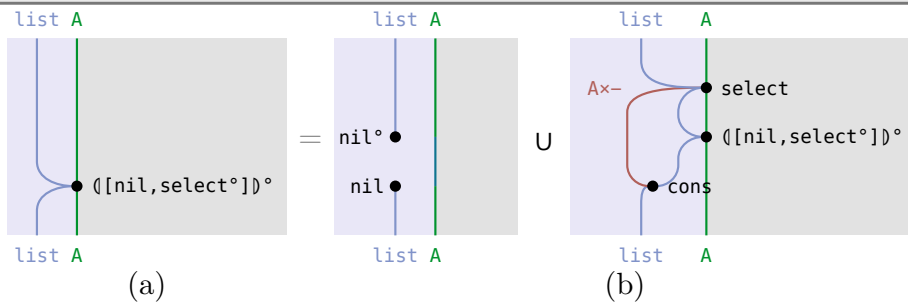
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



|     |                                |                                        |
|-----|--------------------------------|----------------------------------------|
| (a) | ok cons perm                   | cons perm = (1 × perm) cons perm       |
| (b) | ok (1 × perm) cons perm        | Ex 6.22, ok (1 × perm) = (1 × perm) ok |
| (c) | (1 × perm) ok cons perm        | select <sup>°</sup> ok cons perm       |
| (d) | (1 × perm) select <sup>°</sup> |                                        |

$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \cup \text{select} (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

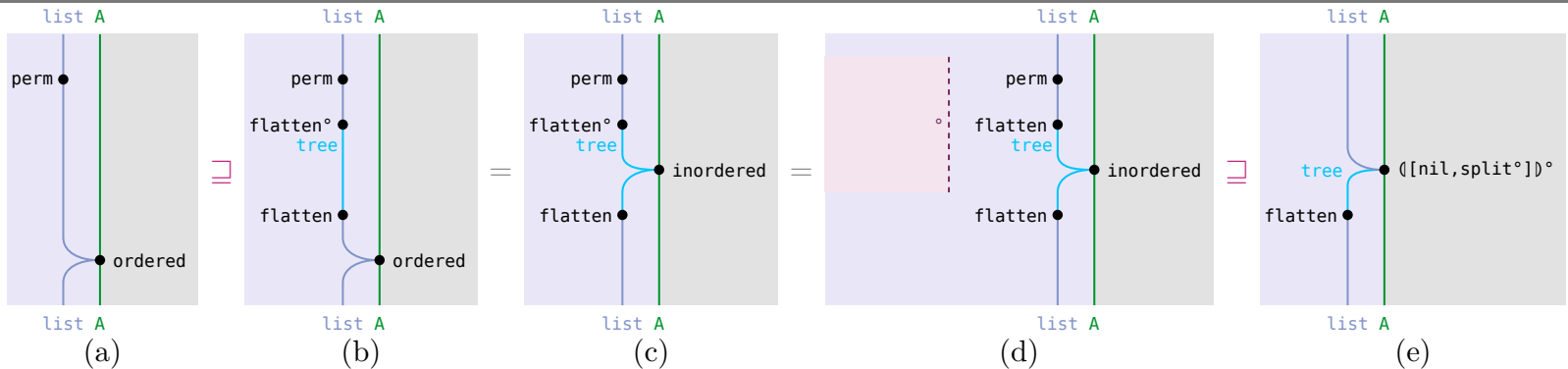
selection sort returns [] on [], and otherwise selects (a,y), sorts y and conses a back on



|     |                                                                                              |                                  |
|-----|----------------------------------------------------------------------------------------------|----------------------------------|
| (a) | $[nil, \text{select}^\circ]^\circ$                                                           | unfolding the converse of a fold |
| (b) | $nil^\circ nil \cup \text{select} (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons}$ |                                  |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \text{check}' \text{ join perm} \Rightarrow ([nil, \text{split}^\circ])^\circ \text{flatten} \text{perm ordered} \quad (6.6e)$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input

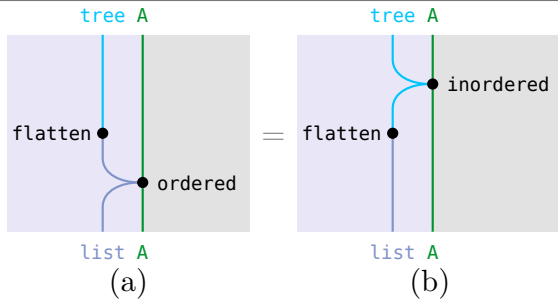


|     |                                                  |                                     |
|-----|--------------------------------------------------|-------------------------------------|
| (a) | perm ordered                                     | flatten is simple                   |
| (b) | perm flatten <sup>°</sup> flatten ordered        | flatten ordered = inordered flatten |
| (c) | perm flatten <sup>°</sup> inordered flatten      | converses                           |
| (d) | (inordered flatten perm) <sup>°</sup> flatten    | fusion (6.4) under (6.6j)           |
| (e) | $[nil, \text{split}^\circ]^\circ \text{flatten}$ |                                     |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten}$$

(6.6f)

flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening

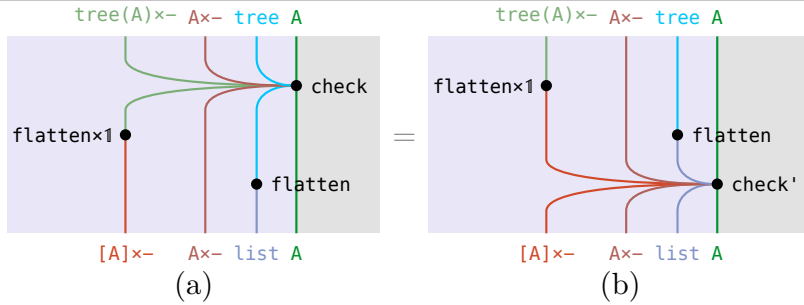


|     |                   |                                                                       |
|-----|-------------------|-----------------------------------------------------------------------|
| (a) | flatten ordered   | R transitive: a flattened tree is ordered iff every node passes check |
| (b) | inordered flatten |                                                                       |

$$\text{check (flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{check'}$$

(6.6g)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening

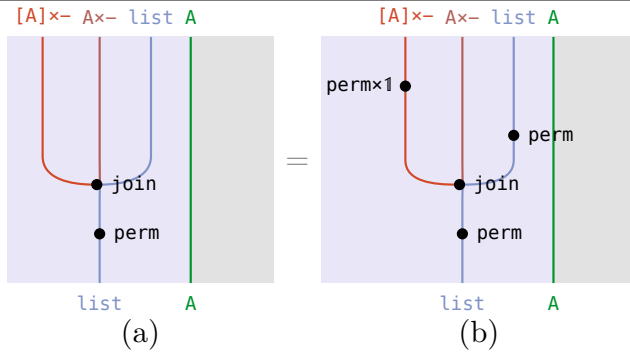


|     |                                  |                                  |
|-----|----------------------------------|----------------------------------|
| (a) | check (flatten × (1 × flatten))  | b intree x ⇔ b inlist flatten(x) |
| (b) | (flatten × (1 × flatten)) check' |                                  |

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{join perm}$$

(6.6h)

permuting the list  $x \mathrel{++} [a] \mathrel{++} y$  is the same as first permuting  $x$  and  $y$  and then permuting the joined list

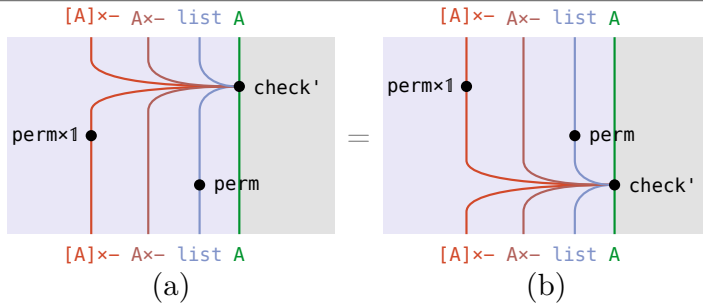


|     |                               |                                        |
|-----|-------------------------------|----------------------------------------|
| (a) | join perm                     | perm is transitive and closed under ++ |
| (b) | (perm × (1 × perm)) join perm |                                        |

$$\text{check' (perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{check'}$$

(6.6i)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

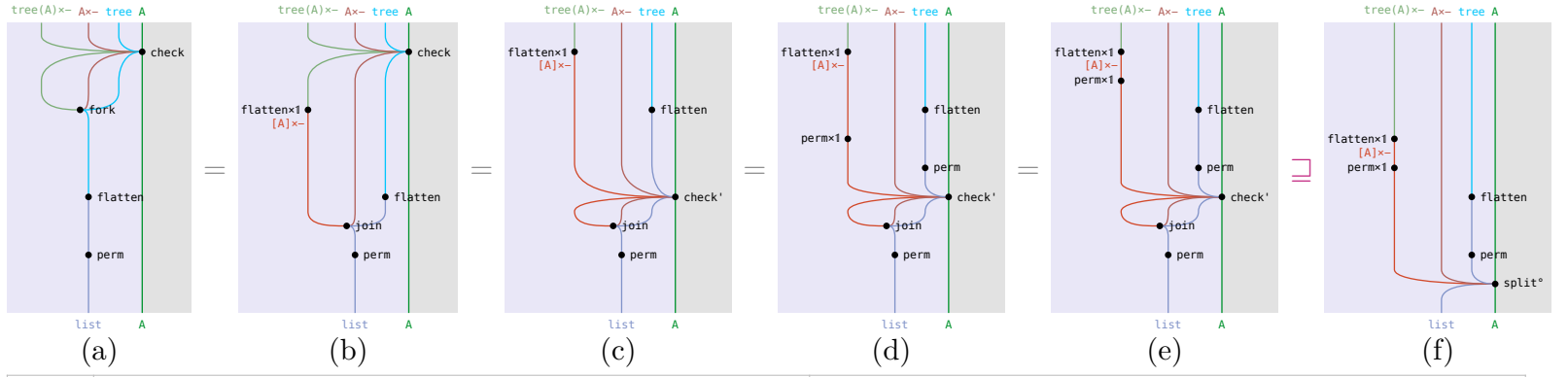


|     |                            |                               |
|-----|----------------------------|-------------------------------|
| (a) | check' (perm × (1 × perm)) | b inlist x ⇔ b inlist perm(x) |
| (b) | (perm × (1 × perm)) check' |                               |



**split° ⊔ check' join perm ⇒ ((flatten perm) × (1 × (flatten perm))) split° ⊔ check fork flatten perm** (6.6j)

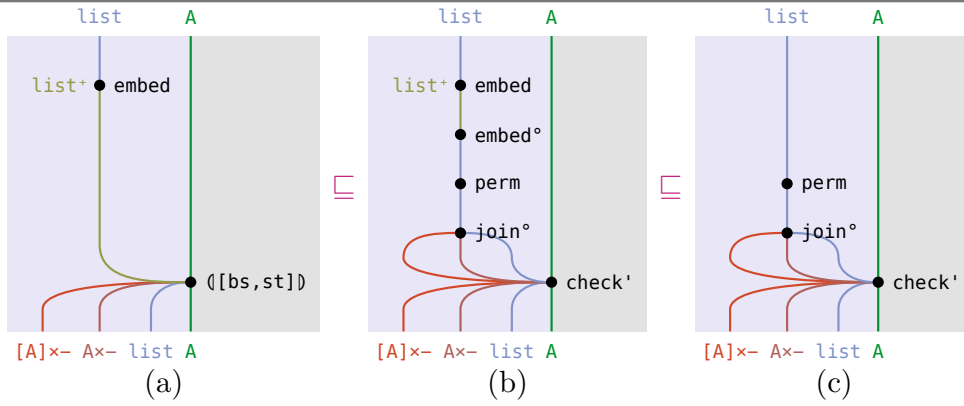
undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



|     |                                                                |                                                                   |
|-----|----------------------------------------------------------------|-------------------------------------------------------------------|
| (a) | check fork flatten perm                                        | fork flatten = F(flatten) join                                    |
| (b) | check (flatten × (1 × flatten)) join perm                      | check F(flatten) = F(flatten) check'                              |
| (c) | (flatten × (1 × flatten)) check' join perm                     | join perm = F(perm) join perm, check'<br>F(perm) = F(perm) check' |
| (d) | (flatten × (1 × flatten)) (perm × (1 × perm)) check' join perm | functors                                                          |
| (e) | ((flatten perm) × (1 × (flatten perm))) check' join perm       | split° ⊔ check' join perm                                         |
| (f) | ((flatten perm) × (1 × (flatten perm))) split°                 |                                                                   |

**bs ⊔ wrap perm join° check' ∧ (1 × (perm join° check')) st ⊔ cons perm join° check' ⇒ embed([bs, st]) ⊔ perm join° check'** (6.6k)

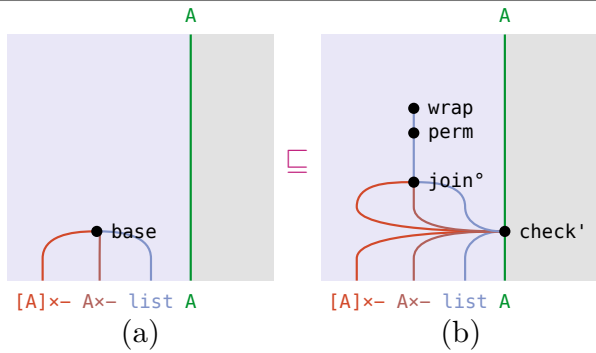
turning a list into a non-empty list and folding it with any **[bs, st]** meeting the two conditions below gives only **checked** triples **(x, a, y)** whose join **x # [a] # y** is a permutation of the list



|     |                                |                                                    |
|-----|--------------------------------|----------------------------------------------------|
| (a) | embed([bs, st])                | (6.6l) and (6.6m), induction on the non-empty list |
| (b) | embed embed° perm join° check' | embed is simple                                    |
| (c) | perm join° check'              |                                                    |

**base ⊔ wrap perm join° check'** (6.6l)

**base(a)** is a **checked** triple whose join is a permutation of the one-element list **[a]**

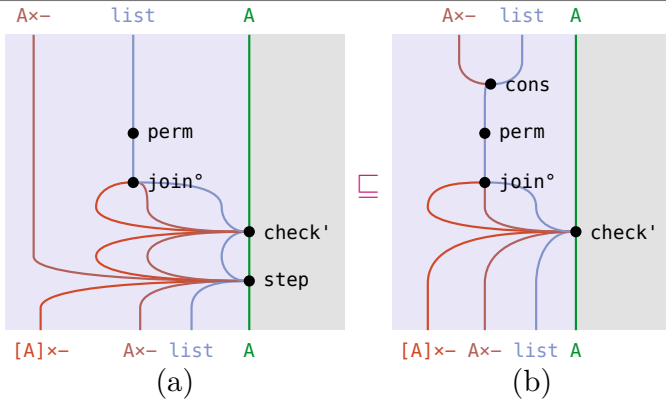


|     |                        |                                                                  |
|-----|------------------------|------------------------------------------------------------------|
| (a) | base                   | $[] \# [a] \# [] = [a]$ , and <b>check'</b> holds on empty lists |
| (b) | wrap perm join° check' |                                                                  |

(6.6m)

$(\forall a. \forall b. \text{leb}(a,b)=\text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b)=\text{false} \Rightarrow R(b,a)) \Rightarrow (1 \times (\text{perm} \text{ join}^\circ \text{ check}')) \text{ step} \sqsubseteq \text{cons perm join}^\circ \text{ check}'$

if  $(x,b,y)$  is a **checked** triple for a permutation of  $\mathbf{l}$ , then  $\text{step}(a, (x,b,y))$  is a **checked** triple for a permutation of  $[a] \# \mathbf{l}$



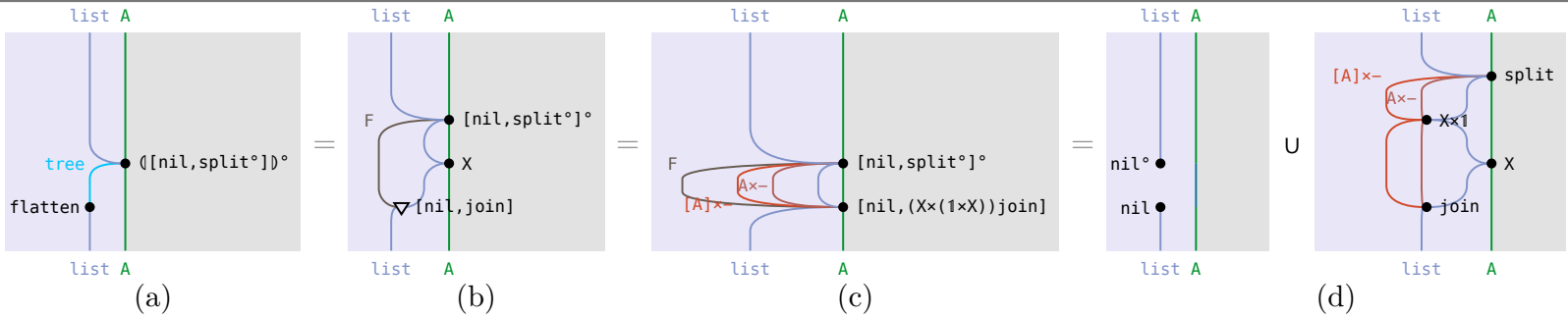
|     |                                                                   |                                                                               |
|-----|-------------------------------------------------------------------|-------------------------------------------------------------------------------|
| (a) | $(1 \times (\text{perm join}^\circ \text{ check}')) \text{ step}$ | $aRb$ puts $a$ in front of $x$ ; otherwise $bRa$ and $a$ goes in front of $y$ |
| (b) | $\text{cons perm join}^\circ \text{ check}'$                      |                                                                               |

(6.6n)

$(\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten} = \text{nil}^\circ \text{ nil} \cup \text{split}$

$((\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten}) \times (1 \times (\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten})) \text{ join}$

quicksort returns  $[]$  on  $[]$ , and otherwise splits into  $(x,a,y)$ , sorts  $x$  and  $y$  and joins them around  $a$

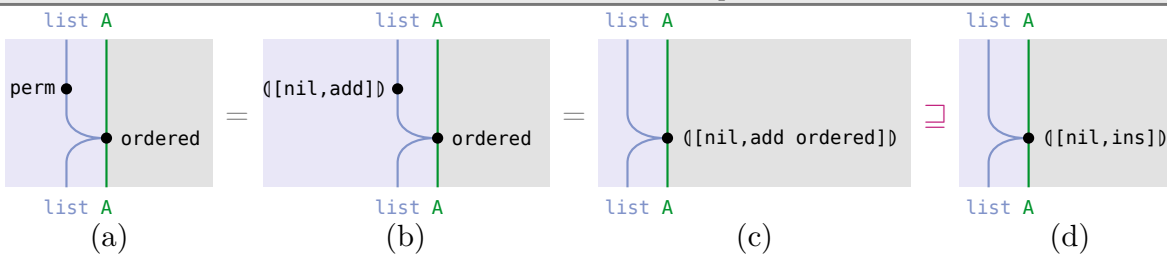


|     |                                                                                             |                                                                                     |
|-----|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| (a) | $(\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten}$              | hylomorphism theorem                                                                |
| (b) | $[\text{nil}, \text{split}^\circ]^\circ F(X) [\text{nil}, \text{join}]$                     | $F(X) [\text{nil}, \text{join}] = [\text{nil}, (X \times 1 \times X) \text{ join}]$ |
| (c) | $[\text{nil}, \text{split}^\circ]^\circ [\text{nil}, (X \times (1 \times X)) \text{ join}]$ | coproduct                                                                           |
| (d) | $\text{nil}^\circ \text{ nil} \cup \text{split } (X \times (1 \times X)) \text{ join}$      |                                                                                     |

(6.6o)

$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow (\llbracket \text{nil}, \text{ins} \rrbracket) \sqsubseteq \text{perm ordered}$

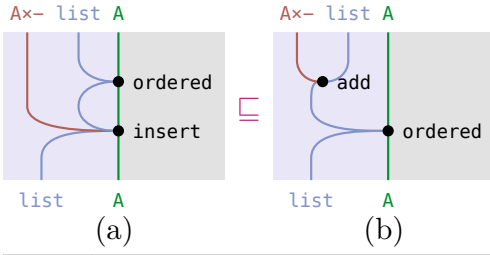
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with  $[\text{nil}, \text{ins}]$  gives only sorted permutations



|     |                                                                  |                                                                              |
|-----|------------------------------------------------------------------|------------------------------------------------------------------------------|
| (a) | $\text{perm ordered}$                                            | $\text{perm} = (\llbracket \text{nil}, \text{add} \rrbracket)$               |
| (b) | $(\llbracket \text{nil}, \text{add} \rrbracket) \text{ ordered}$ | fusion, $\text{add ordered} = (1 \times \text{ordered}) \text{ add ordered}$ |
| (c) | $(\llbracket \text{nil}, \text{add ordered} \rrbracket)$         | $(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$       |
| (d) | $(\llbracket \text{nil}, \text{ins} \rrbracket)$                 |                                                                              |

$$(\forall a. \forall b. \text{leb}(a,b)=\text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b)=\text{false} \Rightarrow R(b,a)) \wedge (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{insert} \sqsubseteq \text{add ordered} \quad (6.6p)$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



|     |                                           |                                                               |
|-----|-------------------------------------------|---------------------------------------------------------------|
| (a) | $(1 \times \text{ordered}) \text{insert}$ | insert splices <b>a</b> in, and <b>R</b> transitive and total |
| (b) | $\text{add ordered}$                      |                                                               |

## §6.7. Closure

$$R^* = (\mu X : 1 \cup XR) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R\***

$$1 \in X \wedge XX \in X \Rightarrow REX \Leftrightarrow R^*EX \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R\***

$$1 \in R^* \quad (6.7c)$$

**R\*** is reflexive

|               |     |               |                      |
|---------------|-----|---------------|----------------------|
|               | (a) | $1$           |                      |
| $\sqsubseteq$ | (b) | $1 \cup RR^*$ | union                |
| $=$           | (c) | $R^*$         | fixed point, Thm 6.1 |

$$R \in R^* \quad (6.7d)$$

**R\*** contains **R**

|               |     |        |                              |
|---------------|-----|--------|------------------------------|
|               | (a) | $R$    |                              |
| $=$           | (b) | $R$    | identity                     |
| $\sqsubseteq$ | (c) | $RR^*$ | $1 \in R^* \text{ — (6.7c)}$ |
| $\sqsubseteq$ | (d) | $R^*$  | fixed point                  |

$$R^*R^* \in R^* \quad (6.7e)$$

**R\*** is transitive, because **R\*/R\*** is a prefixed point of  $X \mapsto 1 \cup RX$

|               |     |                          |                                                                         |
|---------------|-----|--------------------------|-------------------------------------------------------------------------|
|               | (a) | $(1 \cup R(R^*/R^*))R^*$ |                                                                         |
| $=$           | (b) | $R^* \cup R(R^*/R^*)R^*$ | $(P \cup Q)T = PT \cup QT \text{ — composition distributes over } \cup$ |
| $\sqsubseteq$ | (c) | $R^* \cup RR^*$          | $(S/S)S \in S$                                                          |
| $\sqsubseteq$ | (d) | $R^*$                    | $RR^* \in R^* \text{ — (6.7d)}$                                         |

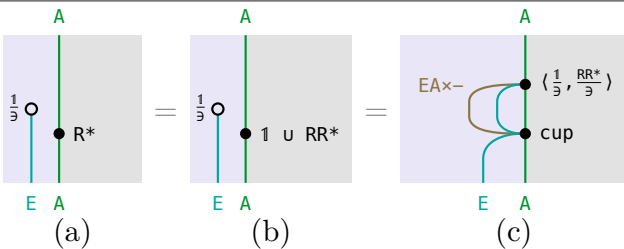
$$1 \in X \wedge XX \in X \wedge REX \Rightarrow R^*EX \quad (6.7f)$$

a reflexive transitive **X** containing **R** is a prefixed point of  $X \mapsto 1 \cup RX$ , so it contains **R\***

|               |     |             |                    |
|---------------|-----|-------------|--------------------|
|               | (a) | $1 \cup RX$ |                    |
| $\sqsubseteq$ | (b) | $X \cup XX$ | $1 \in X, R \in X$ |
| $\sqsubseteq$ | (c) | $X$         | $XX \in X$         |

$$\frac{R^*}{\exists} = \left( \frac{1}{\exists}, \frac{RR^*}{\exists} \right) \text{cup} \quad (6.7g)$$

the set of **R\***-successors of **a** is **a** itself joined with the **R\***-successors of its **R**-successors



|     |                                                                     |                                                         |
|-----|---------------------------------------------------------------------|---------------------------------------------------------|
| (a) | $\frac{R^*}{\exists}$                                               | $R^* = 1 \cup RR^*$                                     |
| (b) | $\frac{1 \cup RR^*}{\exists}$                                       | $\wedge(R \cup S) = (\wedge(R), \wedge(S)) \text{ cup}$ |
| (c) | $\left( \frac{1}{\exists}, \frac{RR^*}{\exists} \right) \text{cup}$ |                                                         |

|                                                             |        |
|-------------------------------------------------------------|--------|
| $R - \emptyset = R$                                         | (6.7h) |
| taking nothing away leaves $R$                              |        |
| $R \cup (S - R) = R \cup S$                                 |        |
| beside $R$ , only the part of $S$ outside $R$ adds anything |        |
| $R - (S \cup T) = R - S - T$                                |        |
| taking away $S$ and then $T$ takes away $S \cup T$          |        |
| $(R \cup S) - T = (R - T) \cup (S - T)$                     |        |
| taking $T$ away from a union takes it away from each part   |        |

|                                                                                                                                   |        |
|-----------------------------------------------------------------------------------------------------------------------------------|--------|
| $\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$ | (6.7i) |
| the least fixed point of $\varphi$ after $\psi$ is $\varphi$ applied to the least fixed point of $\psi$ after $\varphi$           |        |

|                                                                                                               |        |
|---------------------------------------------------------------------------------------------------------------|--------|
| $SR^*=(\mu X : S \cup XR)$                                                                                    | (6.7j) |
| $S$ followed by any number of $R$ steps is the least $X$ containing $S$ and closed under a further $R$ step   |        |
| $R^*S=(\mu X : S \cup RX)$                                                                                    |        |
| any number of $R$ steps followed by $S$ is the least $X$ containing $S$ and closed under an $R$ step in front |        |

| $\theta(\emptyset, S) = SR^*$ (6.7k)                                    |     |                                                    |                              |
|-------------------------------------------------------------------------|-----|----------------------------------------------------|------------------------------|
| started with nothing found and $S$ to explore, $\theta$ computes $SR^*$ |     |                                                    |                              |
|                                                                         | (a) | $\theta(\emptyset, S)$                             |                              |
| =                                                                       | (b) | $\emptyset \cup (\mu X : S \cup (XR - \emptyset))$ | definition of $\theta$       |
| =                                                                       | (c) | $(\mu X : S \cup XR)$                              | $R - \emptyset = R$ — (6.7h) |
| =                                                                       | (d) | $SR^*$                                             | (6.7j)                       |

| $\theta(P, \emptyset) = P$ (6.7l)                                |     |                                            |                                                                |
|------------------------------------------------------------------|-----|--------------------------------------------|----------------------------------------------------------------|
| with nothing left to explore, $\theta$ returns what it has found |     |                                            |                                                                |
|                                                                  | (a) | $\theta(P, \emptyset)$                     |                                                                |
| =                                                                | (b) | $P \cup (\mu X : \emptyset \cup (XR - P))$ | definition of $\theta$                                         |
| =                                                                | (c) | $P \cup \emptyset$                         | $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$ |
| =                                                                | (d) | $P$                                        | $P \cup \emptyset = P$                                         |